members are on leave for higher education in abroad. There are more than 700 undergraduate and more than 30 graduate students in the department.

1.2 Vision and Missions of the Department

1.2.1 Vision

To be a globally recognized leader in Computer Science and Software Engineering through transformative education, impactful research, innovation, ethical practices, and development of society-responsible professionals.

1.2.2 Missions

The missions of the CSE department are:

- **Mission 01 (M1):** To provide high quality education that develops strong foundation in analytical and critical thinking, problem solving abilities, creativity, innovation and professional competence.
- **Mission 02 (M2):** To offer a state-of-the-art curriculum that integrates theory and practice, project-based learning, research and emerging technologies to meet national and global needs for industry and academia.
- **Mission 03 (M3):** To promote excellence in research, innovation, entrepreneurship and knowledge creation through interdisciplinary collaboration among academia, industry, government, and society.
- **Mission 04 (M4):** To cultivate ethical values, leadership qualities, environmental consciousness, social responsibilities, and life-long learning so that graduates can contribute positively, to humanity, sustainable development, and welfare of continuously changing technology-driven society.

1.3 Programs Offered by the Department

1. Doctor of Philosophy in Computer Science and Engineering, Ph.D. (CSE)
2. Master of Science in Computer Science and Engineering, M.Sc. Engg. (CSE)
3. Master of Engineering in Computer Science and Engineering, M. Engg. (CSE)
4. Bachelor of Science in Computer Science and Engineering, B.Sc. Engg. (CSE)
5. Bachelor of Science in Software Engineering, B.Sc. (SWE)

Chapter 2

Program Information

In this era, software is crucial to the operation of computers. It has real-life implications in many industries — including medicine, communications, business, military, aerospace, scientific, and general computing. Using principles and techniques of computer science, engineering, and mathematical analysis, software engineers empower computers with innovative applications to perform tasks smarter, faster, and better. Therefore, IUT started to offer a Bachelor of Science in Software Engineering under the Computer Science and Engineering department in the academic year of 2017–18.

Through this program, IUT promises to provide students with a strong foundation in software engineering using a combination of Classroom Study, Laboratory Sessions, Software Project Labs, and Design Projects. The program blends engineering principles, computing skills, project leadership, and software construction to equip students with a comprehensive understanding of the field and to prepare graduates for the workforce or future study.

The program is designed around a set of core courses that introduces the fundamentals of software engineering, followed by a broader range of courses. Students could choose to augment their core with more Software Systems and Security oriented courses (e.g., Software Environments, Security Risk Analysis and Management), Data Science courses (e.g., Data Mining, Big Data and Large-scale Computing,), Web Services and Applications oriented courses (e.g., Web Programming, User Interface Design and Evaluation), or Graphics and Game related courses (e.g., Animation for Computer Games, Artificial Intelligence for Games). Each of these areas is covered by a dedicated set of core and extended courses. In short, by providing a careful balance between theory and practice, the program will prepare students for central software positions in industry, government organizations, and institutions where software engineering has become a key activity.

The unique characteristics of the newly introduced software engineering program can